

5 — Results

1 Experiments

Run matrix

Six tasks: 5-way classification, and parameter estimation for each of the five families. Each run through both arms at every feature setting.

- **PH arm**, 18 settings: 6 filtrations (Rips, alpha, DTM_{5,10,15,20}) $\times$ $\{H_0, H_1, H_0+H_1\}$
- **Classical arm**, 4 settings:
 - L on **fixed** alone (VIHRS feature set)
 - L, F, G, J all on **fixed**, stacked
 - L, F, G, J all on **sqrtn**, stacked
 - L on **fixed**, F, G, J on **sqrtn** (per-function grids, one encoder each)
- $6 \times (18 + 4) = 132$ configurations $\times$ 10 seeds = 1320 runs, all complete

Conduct

- Everything is evaluated on the same 2000 test θ per family, fixed before any run and untouched until evaluation
- Best epoch ranges 7–196 with median 34. Six runs of 1320 reached the 200-epoch cap, all of them Matérn parameters and five of them on a 1-D H_0 input
- Seed sd of a marginal cell estimate is 0.001–0.004 for accuracy, an order of magnitude below the θ bootstrap width

Regime coordinate

Cells are 8 log- $\tilde{\delta}$ bins on $[0.1, 4]$ and 3 log- $\bar{n}$ bins on $[100, 800]$, edges from the prior and shared by every family and method. Binning is on $\tilde{\delta}$ *recomputed under the current null tables*, whose radius set keeps radii with at least $p = 5$ expected CSR pairs. The prior was swept against the earlier $p = 10$ tables, so the label of a fixed θ has moved, and by different amounts per family:

test θ per $\tilde{\delta}$ bin	0.10	0.16	0.25	0.40	0.63	1.00	1.59	2.52	$\tilde{\delta}_5/\tilde{\delta}_{10}$
Thomas	244	259	247	235	271	243	266	230	0.98
nested	270	227	272	272	263	245	239	209	0.99
LGCP	223	281	272	229	279	227	258	225	0.99
Matérn	73	232	208	230	175	262	588	232	1.40
all four	810	999	999	966	988	977	1351	896	

- For the three clustering families the relabelling is a uniform -1 – 2% and the bins stay flat at ≈ 250 θ
- Matérn moves: median ratio 1.40, peaking at 1.86 near $\tilde{\delta}_{10} \approx 0.8$ and back to 0.99 in the top two bins. A hard core of radius $R < r_{\min}$ enters $\tilde{\delta}$ only through the pair deficit it leaves *above* $r_{\min}$, diluted by the CSR count there (01, 02); dropping p to 5 extends $R(n)$ below R and the deficit is seen directly. At large R the core is already inside $R(n)$ at either cutoff, so the label does not move

- Consequence: Matérn’s prior is no longer flat across $\tilde{\delta}$ bins. Cell estimates are unaffected — each is read within its own bin — but the Matérn intervals are wide at $\tilde{\delta} = 0.10$ (73θ) and tight at 1.59 (588θ)
- The sweep itself is unchanged. The shape-rule floor in `configs/simulation` stays at 10 expected pairs: that is a choice about which scales are worth sweeping, deliberately not the test’s cutoff
- Poisson has $\tilde{\delta} = 0$ and falls in no $\tilde{\delta}$ bin. It appears in the marginal and $\bar{n}$ rows only, so a $\tilde{\delta}$ cell of the classification task holds four families and chance in it is 0.25

2 Classification

Accuracy on the whole test prior, mean over 10 seeds (chance 0.2; bootstrap half-widths ≈ 0.008):

	Rips	alpha	DTM ₅	DTM ₁₀	DTM ₁₅	DTM ₂₀
H_0	0.494	0.493	0.470	0.477	0.476	0.473
H_1	0.338	0.362	0.317	0.319	0.320	0.319
H_0+H_1	0.489	0.495	0.471	0.472	0.472	0.468

Classical arm:

L fixed (VIHRS set)	0.518
L, F, G, J fixed	0.539
L, F, G, J <code>sqrtn</code>	0.540
L, F, G, J per-function grids	0.535

Paired against the best classical setting (L, F, G, J `sqrtn`), same θ resamples:

- L alone: -0.023 $[-0.027, -0.019]$
- alpha H_0+H_1 : -0.046 $[-0.050, -0.041]$
- DTM₁₀ H_0 : -0.064 $[-0.069, -0.059]$
- Adding H_1 to H_0 : -0.005 to $+0.002$, negative for four of six filtrations

By regime

$\tilde{\delta}$ cells hold four families (Poisson has $\tilde{\delta} = 0$), so chance in them is 0.25; $\bar{n}$ cells hold five, so chance is 0.2.

accuracy by $\tilde{\delta}$	0.10	0.16	0.25	0.40	0.63	1.00	1.59	2.52
L, F, G, J <code>sqrtn</code>	0.101	0.189	0.312	0.439	0.486	0.627	0.783	0.780
L fixed	0.106	0.201	0.308	0.428	0.468	0.602	0.731	0.673
Rips H_0	0.088	0.165	0.244	0.380	0.438	0.567	0.722	0.683
alpha H_0	0.086	0.167	0.241	0.377	0.436	0.572	0.722	0.678
DTM ₁₀ H_0	0.093	0.143	0.200	0.323	0.429	0.578	0.731	0.721
DTM ₁₀ H_1	0.113	0.179	0.190	0.233	0.274	0.358	0.473	0.522

	accuracy by $\bar{n}$			recall by family (all regimes)				
	100	200	400	Pois.	Thomas	nested	Matérn	LGCP
L, F, G, J <code>sqrtn</code>	0.544	0.544	0.533	0.773	0.213	0.426	0.903	0.386
L fixed	0.518	0.518	0.518	0.764	0.214	0.415	0.908	0.288
Rips H_0	0.508	0.499	0.475	0.760	0.210	0.340	0.845	0.316
alpha H_0	0.509	0.498	0.472	0.758	0.200	0.339	0.848	0.321
DTM ₁₀ H_0	0.495	0.483	0.451	0.707	0.198	0.419	0.714	0.346
DTM ₁₀ H_1	0.334	0.314	0.310	0.390	0.116	0.237	0.556	0.297

Regime Breakdown

The marginal recalls above split by $\tilde{\delta}$. Restricted to one family, accuracy is that family's recall, so these are the same cells as the table above, cut by truth rather than pooled. Three settings: the best classical, the best PH (H_0 on a geometric filtration), and DTM₁₀ H_0 , the density-field filtration.

	recall by $\tilde{\delta}$	0.10	0.16	0.25	0.40	0.63	1.00	1.59	2.52
Thomas	L, F, G, J sqrtn	0.027	0.033	0.062	0.080	0.156	0.286	0.466	0.615
	Rips H_0	0.043	0.048	0.070	0.106	0.190	0.273	0.424	0.536
	DTM ₁₀ H_0	0.037	0.058	0.086	0.109	0.170	0.271	0.391	0.470
nested	L, F, G, J sqrtn	0.110	0.139	0.231	0.403	0.551	0.635	0.697	0.727
	Rips H_0	0.080	0.109	0.162	0.279	0.416	0.494	0.560	0.717
	DTM ₁₀ H_0	0.116	0.145	0.219	0.356	0.532	0.653	0.672	0.746
Matérn	L, F, G, J sqrtn	0.351	0.500	0.872	0.982	1.000	1.000	1.000	1.000
	Rips H_0	0.282	0.369	0.619	0.883	0.969	1.000	1.000	1.000
	DTM ₁₀ H_0	0.210	0.270	0.312	0.537	0.706	0.863	0.986	1.000
LGCP	L, F, G, J sqrtn	0.090	0.116	0.193	0.304	0.424	0.554	0.697	0.773
	Rips H_0	0.082	0.150	0.198	0.276	0.365	0.462	0.544	0.474
	DTM ₁₀ H_0	0.089	0.115	0.198	0.290	0.408	0.496	0.556	0.666

The same recalls by $\bar{n}$ (Poisson appears here, having no $\tilde{\delta}$ bin):

	L, F, G, J sqrtn			Rips H_0			DTM ₁₀ H_0		
$\bar{n}$	100	200	400	100	200	400	100	200	400
Poisson	0.781	0.778	0.762	0.744	0.779	0.759	0.707	0.716	0.698
Thomas	0.261	0.217	0.161	0.247	0.207	0.175	0.234	0.203	0.155
nested	0.401	0.431	0.447	0.309	0.343	0.367	0.395	0.431	0.430
Matérn	0.886	0.905	0.918	0.897	0.855	0.780	0.777	0.714	0.647
LGCP	0.379	0.391	0.389	0.332	0.313	0.304	0.351	0.359	0.329

3 Parameters

RMSE in log units, pooled over a family's targets, over the whole test prior:

	Poisson	Thomas	nested	Matérn	LGCP
L, F, G, J fixed	0.063	0.878	0.593	0.118	0.627
L, F, G, J sqrtn	0.063	0.896	0.593	0.127	0.631
L fixed	0.063	0.890	0.596	0.117	0.632
Rips H_0	0.063	1.021	0.605	0.101	0.671
alpha H_0	0.063	1.020	0.605	0.102	0.672
alpha H_0+H_1	0.063	0.946	0.602	0.107	0.668
DTM ₁₀ H_0	0.063	0.945	0.606	0.158	0.682
DTM ₁₀ H_1	0.063	1.019	0.684	0.319	0.873

Regime Breakdown

Pooled log-RMSE by $\tilde{\delta}$: the best classical setting (L, F, G, J fixed) against the family's best PH setting, and their paired difference on the same θ resamples. Positive means PH is worse; * marks a 95% interval that excludes 0.

	pooled log-RMSE by $\tilde{\delta}$	0.10	0.16	0.25	0.40	0.63	1.00	1.59	2.52
Thomas	L, F, G, J fixed	1.215	1.117	0.919	0.880	0.915	0.676	0.525	0.459
	DTM ₅ H_0+H_1	1.230	1.140	0.969	0.932	1.011	0.730	0.618	0.487
	paired	+0.015	+0.023	+0.050*	+0.052*	+0.095*	+0.054*	+0.093*	+0.028
nested	L, F, G, J fixed	0.732	0.641	0.605	0.619	0.596	0.568	0.450	0.433
	alpha H_0+H_1	0.754	0.662	0.612	0.628	0.605	0.592	0.445	0.398
	paired	+0.022*	+0.021*	+0.008	+0.009	+0.010	+0.024*	−0.006	−0.035*
Matérn	L, F, G, J fixed	0.170	0.134	0.105	0.119	0.131	0.118	0.100	0.120
	Rips H_0	0.181	0.146	0.112	0.120	0.108	0.085	0.064	0.054
	paired	+0.010*	+0.012*	+0.007*	+0.001	−0.024*	−0.033*	−0.036*	−0.066*
LGCP	L, F, G, J fixed	0.888	0.800	0.724	0.640	0.576	0.448	0.352	0.297
	alpha H_0+H_1	0.917	0.831	0.743	0.685	0.622	0.543	0.421	0.339
	paired	+0.029*	+0.031*	+0.020*	+0.045*	+0.046*	+0.094*	+0.068*	+0.042*

- Thomas, nested and LGCP improve with $\tilde{\delta}$ in both arms. LGCP improves strictly, while the two clustering families plateau at $\tilde{\delta} \approx 0.4$ –0.6
- On Matérn the classical arm is flat over the whole range (0.105–0.170, no trend above $\tilde{\delta} \approx 0.25$) while Rips H_0 falls 0.181 $\rightarrow$ 0.054
- Classical arm leads everywhere except Matérn above $\tilde{\delta} \approx 0.4$ and nested’s top two bins. On nested, PH is ahead by 0.006 at $\tilde{\delta} = 1.59$ and by 0.035 at 2.52; only the second paired interval clears 0, so the top bin is the one win that is established
- Only Matérn’s gap widens with $\tilde{\delta}$. LGCP’s and Thomas’s are widest in the middle and close again at the extremes, where both arms are either uninformative or saturated.

The same, pooled by $\bar{n}$:

	L, F, G, J fixed			Rips H_0			DTM ₁₀ H_0		
$\bar{n}$	100	200	400	100	200	400	100	200	400
Poisson	0.079	0.061	0.043	0.079	0.062	0.043	0.080	0.062	0.043
Thomas	0.791	0.870	0.967	0.882	1.010	1.155	0.836	0.925	1.062
nested	0.520	0.586	0.660	0.521	0.602	0.678	0.534	0.605	0.668
Matérn	0.132	0.112	0.108	0.103	0.094	0.105	0.159	0.150	0.165
LGCP	0.537	0.631	0.698	0.560	0.673	0.758	0.575	0.685	0.766

More points make every shape parameter *harder*, in both arms: at fixed $\tilde{\delta}$ a larger $\bar{n}$ means a weaker per-point departure, since $\tilde{\delta}$ is defined at $\bar{n}$ and the draw is independent of it. Poisson is the exception — its target is $\bar{n}$ itself, and 0.079/0.061/0.043 tracks $1/\sqrt{\bar{n}}$ (0.084/0.059/0.042 at the bins’ geometric centres), the MLE floor for $n \mid \bar{n}$, reached by all 22 inputs.

Matérn R by regime

The one target where the arms separate, at target rather than pooled resolution. Log-RMSE by $\tilde{\delta}$ bin, and the paired difference of Rips H_0 against L, F, G, J fixed on the same θ resamples:

$\tilde{\delta}$	0.10	0.16	0.25	0.40	0.63	1.00	1.59	2.52
L, F, G, J fixed	0.230	0.177	0.133	0.156	0.174	0.156	0.126	0.147
L fixed	0.241	0.184	0.130	0.150	0.160	0.147	0.124	0.153
alpha H_0	0.249	0.197	0.145	0.158	0.142	0.108	0.066	0.047
Rips H_0	0.246	0.197	0.145	0.157	0.138	0.105	0.066	0.040
DTM ₁₀ H_0	0.386	0.305	0.190	0.203	0.221	0.218	0.173	0.113
paired, Rips H_0	+0.016	+0.020	+0.012	+0.000	-0.037	-0.051	-0.060	-0.107
lower	+0.005	+0.013	+0.004	-0.007	-0.046	-0.057	-0.063	-0.112
upper	+0.027	+0.027	+0.019	+0.007	-0.027	-0.044	-0.056	-0.102

4 Observations

- **PH’s edge is Matérn’s hard core**, above $\tilde{\delta} \approx 0.4$ only. Paired difference on R changes sign there, interval excludes 0; top bin 0.040 vs 0.147 (3.7 $\times$). A forbidden spacing is a length H_0 records and $\hat{K}$ averages over
- **Estimation, not identification.** The classical arm recognises Matérn earlier (recall 1.0 by $\tilde{\delta} \approx 0.6$, PH by ≈ 1.0) and stays ahead on R until $\tilde{\delta} \approx 0.4$; PH only overtakes where the family is already certain
- **Classification goes to the classical arm:** -0.046 paired at best, ahead in every $\tilde{\delta}$ bin from 0.16. With L alone (VIHRS) the gap is -0.023 , mostly LGCP recall ($0.288 \rightarrow 0.386$); the rest comes from F, G, J
- **Where the classification gap lives.** Against Rips H_0 per family: nested in the middle (0.403 vs 0.279 at $\tilde{\delta} = 0.40$), Matérn below saturation (0.872 vs 0.619 at 0.25), LGCP at the top (0.773 vs 0.474). Thomas is level or behind until $\tilde{\delta} \approx 1$, Poisson level throughout. Thomas below $\tilde{\delta} \approx 0.6$ is where every method fails — recall 0.03 – 0.19 against 0.25 chance, and 56 – 71% of those patterns are called Poisson
- **H_1 does not pay.** At most $+0.002$ over H_0 , weakest input alone for every family and task; in 2-D the loops are the complement of clusters already in H_0
- **Geometry vs density.** Rips/alpha beat DTM on classification (0.495 vs 0.477) and Matérn; DTM wins only LGCP recall (0.386 vs 0.321), the one density-field departure. By regime the split is sharper: DTM₁₀ leads Rips on nested in every $\tilde{\delta}$ bin and on LGCP from $\tilde{\delta} \approx 0.25$ up, and trails it on Matérn in every bin below saturation
- **PH degrades with $\bar{n}$, classical does not** ($0.509 \rightarrow 0.472$ vs $0.544 \rightarrow 0.533$): $\sqrt{\bar{n}}$ rescaling makes persistence images $\bar{n}$ -invariant in geometry, not information. The per-family split locates it — Matérn recall goes $0.897 \rightarrow 0.780$ under Rips H_0 while the classical arm goes $0.886 \rightarrow 0.918$. Thomas falls in both arms and nested rises in both, so neither contributes to the gap
- **$\bar{n}$ hurts estimation in both arms.** Pooled log-RMSE rises with $\bar{n}$ for every clustering family and both arms (Thomas $0.79 \rightarrow 0.97$ classical, $0.88 \rightarrow 1.16$ Rips H_0): at fixed $\tilde{\delta}$ more points means a weaker per-point departure. Poisson inverts it because its target is $\bar{n}$
- **Low- $\tilde{\delta}$ bins are below chance** (0.25 for four families): the modal prediction is Poisson (68% of the bottom bin, 59% of the next), excluded there by construction — a property of the prior, not the methods
- **Some targets carry no feature information:** Poisson $\bar{n}$ is 0.063 for all 22 inputs ($\log n$ solves it); nested μ_1 stays at its prior sd (≈ 0.397); Thomas κ and μ agree within 0.003 , being one target given $\kappa = \bar{n}/\mu$
- **The $p = 10 \rightarrow 5$ relabelling moves Matérn only** ($\tilde{\delta}$ up $1.4\times$ at the median, $1.9\times$ near $\tilde{\delta} \approx 0.8$; the other three families within 2%), because a sub- $r_{\min}$ hard core is the one departure the cutoff hides. The Matérn $\tilde{\delta}$ curves above are therefore read on an unevenly populated prior, 73 to 588 test

θ per bin